Unsupervised Learning Project: Customer Segmentation using Clustering

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Contents

1 Introduction

In this project, we apply unsupervised learning techniques to perform customer segmentation using retail transactional data.

Unlike supervised learning, unsupervised learning does not rely on predefined target labels. Instead, its objective is to discover hidden structures and patterns within the dataset.

Customer segmentation is an important task in data analysis and business intelligence because it helps companies better understand customer behavior and improve marketing strategies.

The dataset used in this project contains transactional information from an online retail company, including product purchases, quantities, prices, invoice numbers, and customer identifiers.

Two clustering algorithms are implemented and compared in this study:

- K-Means Clustering
- Hierarchical Clustering

The objective of this project is not only to group customers into meaningful clusters, but also to analyze customer purchasing behavior and compare different clustering approaches.

2 Dataset Description

The dataset used in this project is the Online Retail dataset, which contains transactional records from a UK-based online retail company.

Each row represents a transaction associated with a customer purchase.

The dataset contains several variables such as:

- Invoice Number
- Product Code
- Product Description
- Quantity
- Invoice Date
- Unit Price

- Customer ID
- Country

For the purpose of clustering analysis, only the most relevant variables related to customer purchasing behavior are selected.

The following variables are mainly used to create customer features:

- Quantity
- Unit Price
- Customer ID

Additional customer-level features are later generated during preprocessing.

3 Objective of the Study

The main objective of this project is to segment customers into different groups according to their purchasing behavior using unsupervised learning techniques.

More specifically, the goals are:

- Clean and preprocess the retail dataset
- Create customer behavioral features
- Normalize numerical variables
- Apply clustering algorithms
- Visualize customer groups
- Compare the performance and interpretation of clustering methods

This project also aims to demonstrate the importance of preprocessing and distance-based analysis in clustering problems.

4 Methodology

4.1 Dataset Overview

Before applying clustering algorithms, it is important to understand the structure of the dataset.

The following figure presents a preview of the dataset, including the main variables and transaction format.

```
 $ UnitPrice  : num [1:541909] 2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 1.69 ...
 $ CustomerID : num [1:541909] 17850 17850 17850 17850 17850 ...
 $ Country    : chr [1:541909] "United Kingdom" "United Kingdom" "United Kingdom"
"United Kingdom" ...
     InvoiceNo          StockCode          Description          Quantity
InvoiceDate
 Length    :541909   Length    :541909   Length    :541909   Min.   :-80995.000   Min.
:2010-12-01 08:26:00
 N.unique : 25900    N.unique :   4070   N.unique :   4211   1st Qu.:     1.000   1st
Qu.:2011-03-28 11:34:00
 N.blank  :      0   N.blank  :      0   N.blank  :      0   Median :     3.000   Median
:2011-07-19 17:17:00
 Min.nchar:      6   Min.nchar:      1   Min.nchar:      1   Mean   :     9.552   Mean
:2011-07-04 13:34:57
 Max.nchar:      7   Max.nchar:     12   Max.nchar:     35   3rd Qu.:    10.000   3rd
Qu.:2011-10-19 11:27:00
                                         NAs      :   1454   Max.   : 80995.000   Max.
:2011-12-09 12:50:00

   UnitPrice            CustomerID          Country
 Min.   :-11062.060   Min.   :12346    Length    :541909
 1st Qu.:     1.250   1st Qu.:13953    N.unique :     38
 Median :     2.080   Median :15152    N.blank  :      0
 Mean   :     4.611   Mean   :15288    Min.nchar:      3
 3rd Qu.:     4.130   3rd Qu.:16791    Max.nchar:     20
 Max.   : 38970.000   Max.   :18287
                      NAs    :135080
```

Figure 1: Preview of the Online Retail dataset

The dataset contains both numerical and categorical variables. However, clustering algorithms mainly rely on numerical variables and distance calculations.

4.2 Exploratory Data Analysis (EDA)

Before applying clustering algorithms, an exploratory data analysis was performed to better understand the purchasing behavior of customers in the Online Retail dataset.

The original dataset contains transactional records from an online retail company, where each observation represents a customer purchase. After data cleaning and feature engineering, transaction-level data were aggregated at the customer level in order to create behavioral features.

The main customer features analyzed are:

- Total spending: the total amount spent by each customer
- Purchase frequency: the number of purchases made by each customer
- Average spending: the average amount spent per purchase

These features provide a numerical representation of customer behavior and are used as inputs for the clustering algorithms.

4.2.1 Distribution of Total Spending

The distribution of total spending shows differences in customer purchasing activity. The analysis highlights that customer spending is not uniformly distributed.

A limited number of customers have significantly higher spending values, while most customers present lower spending levels. These variations are important for customer segmentation because clustering algorithms are based on distance measurements and can be influenced by extreme values.

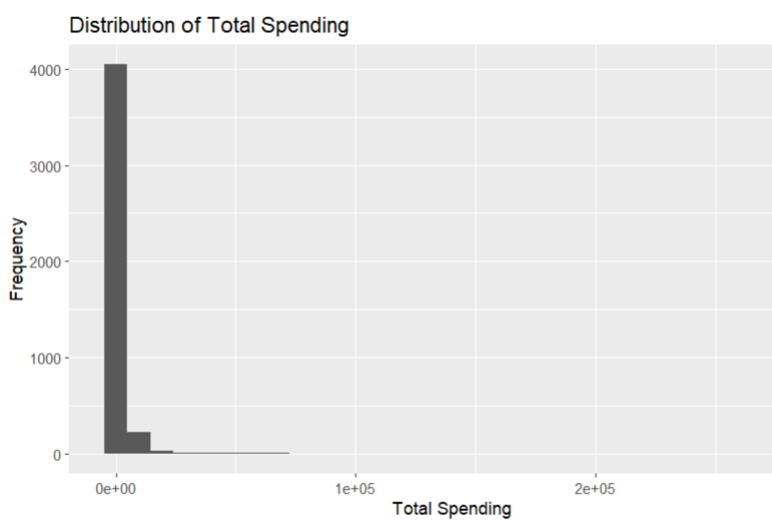

Figure 2: Distribution of total spending

4.2.2 Distribution of Purchase Frequency

Purchase frequency represents the number of transactions performed by each customer.

The distribution shows the presence of different types of customers, ranging from occasional buyers to customers with frequent purchasing activity. This difference in behavior supports the objective of applying clustering methods to identify meaningful customer segments.

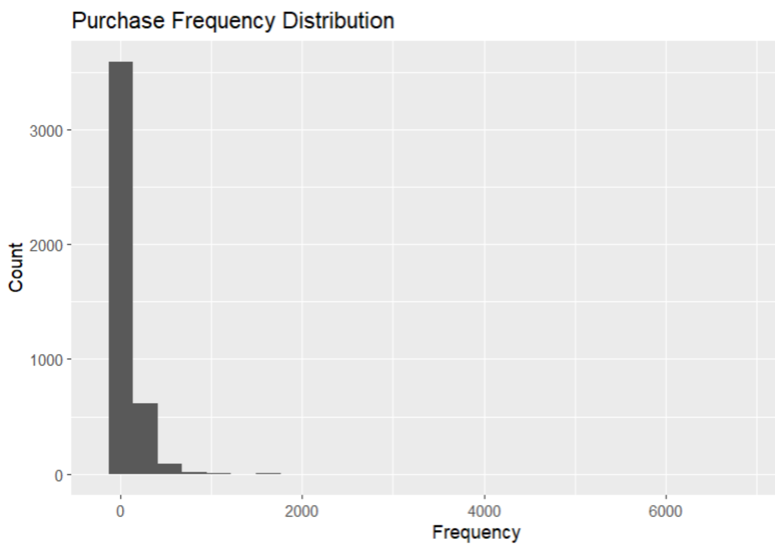

Figure 3: Distribution of purchase frequency

4.3 Data Cleaning

Data preprocessing is an essential step before applying clustering techniques.

First, canceled transactions are removed from the dataset. Transactions whose invoice numbers begin with the letter "C" correspond to canceled orders.

In addition, observations with negative or zero quantities are removed in order to keep only valid purchases.

These preprocessing steps improve the quality and reliability of the clustering analysis.

```
   InvoiceNo           StockCode          Description          Quantity           InvoiceDate
 Length    :397924   Length    :397924   Length    :397924   Min.   :     1.00   Min.   :2010-12-01 08:26:00
 N.unique : 18536   N.unique :   3665   N.unique :   3866   1st Qu.:     2.00   1st Qu.:2011-04-07 11:12:00
 N.blank  :      0   N.blank  :      0   N.blank  :      0   Median :     6.00   Median :2011-07-31 14:39:00
 Min.nchar:      6   Min.nchar:      1   Min.nchar:      6   Mean   :    13.02   Mean   :2011-07-10 23:43:36
 Max.nchar:      6   Max.nchar:     12   Max.nchar:     35   3rd Qu.:    12.00   3rd Qu.:2011-10-20 14:33:00
                                                             Max.   :80995.00   Max.   :2011-12-09 12:50:00
   UnitPrice          CustomerID          Country            TotalPrice
 Min.   :   0.000   Min.   :12346   Length    :397924   Min.   :     0.00
 1st Qu.:   1.250   1st Qu.:13969   N.unique :     37   1st Qu.:     4.68
 Median :   1.950   Median :15159   N.blank  :      0   Median :    11.80
 Mean   :   3.116   Mean   :15294   Min.nchar:      3   Mean   :    22.39
 3rd Qu.:   3.750   3rd Qu.:16795   Max.nchar:     20   3rd Qu.:    19.80
 Max.   :8142.750   Max.   :18287                       Max.   :168469.60
```

Figure 4: Data cleaning process

4.4 Feature Engineering

To perform customer segmentation, new customer-level variables are created from the transactional dataset.

First, the total amount spent for each transaction is calculated using the following formula:

$$\text{TotalAmount} = \text{Quantity} \times \text{UnitPrice}$$

This variable represents the monetary value of each transaction.

Then, the dataset is aggregated by customer in order to generate behavioral indicators such as:

- Total spending
- Purchase frequency
- Average spending

These features allow each customer to be represented numerically for clustering analysis.

A tibble: 6 × 4

CustomerID <dbl>	total_spent <dbl>	frequency <int>	avg_spent <dbl>
12346	77183.60	1	77183.60000
12347	4310.00	182	23.68132
12348	1797.24	31	57.97548
12349	1757.55	73	24.07603
12350	334.40	17	19.67059
12352	2506.04	85	29.48282

6 rows

Figure 5: Customer-level features generated after aggregation

4.5 Data Normalization

Normalization is particularly important for clustering algorithms because they rely on distance calculations.

Variables with larger scales may dominate the clustering process if normalization is not applied.

To avoid this issue, feature scaling is performed using the `scale()` function.

The following transformation standardizes the variables:

$$x' = \frac{x - \mu}{\sigma}$$

where:

- x: original value
- μ: mean of the variable
- σ: standard deviation

The CustomerID column is excluded because it represents an identifier rather than a meaningful analytical variable.

```
      total_spent   frequency    avg_spent
[1,]  8.35867054 -0.39646644 52.539965535
[2,]  0.25101743  0.39464219 -0.030401958
[3,] -0.02854316 -0.26534347 -0.007036759
[4,] -0.03295893 -0.08177130 -0.030133037
[5,] -0.19129345 -0.32653419 -0.033134537
[6,]  0.05031536 -0.02932211 -0.026449294
```

Figure 6: Data normalization process

Normalization ensures that all variables contribute equally to distance calculations.

4.6 Workflow Overview

The following figure summarizes the overall methodology used in this project.

It includes data cleaning, feature engineering, normalization, clustering model training, and cluster visualization.

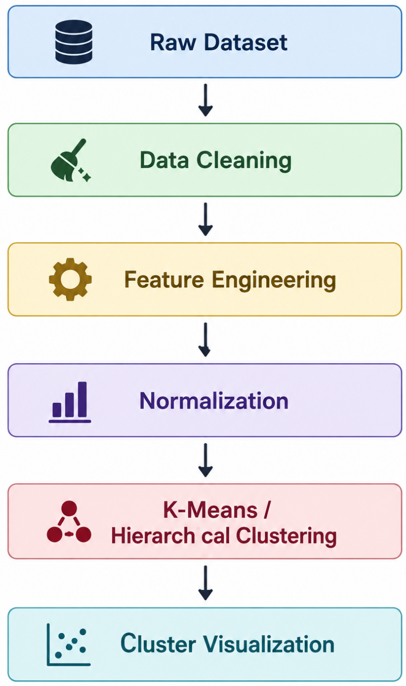

Figure 7: Overview of the clustering workflow

4.7 Distance-Based Clustering

Both clustering algorithms used in this project rely on distance calculations to measure similarity between customers.

Customers with similar purchasing behaviors are grouped together, while customers with different behaviors are placed into separate clusters.

The Euclidean distance is used in this study and is defined as:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

This distance metric measures the similarity between customers using their numerical behavioral features.

4.8 Cluster Evaluation Strategy

Unlike supervised learning, clustering does not use target labels for evaluation.

Therefore, cluster quality is analyzed using:

- Cluster visualization
- Elbow Method
- Dendrogram analysis
- Customer behavior interpretation

These techniques help determine the optimal number of clusters and evaluate the coherence of the segmentation.

5 Clustering Models and Results

5.1 K-Means Clustering

K-Means is one of the most widely used clustering algorithms.

The objective of K-Means is to partition the dataset into k clusters by minimizing the distance between observations and their cluster centroids.

The algorithm works iteratively:

- Initialize cluster centroids
- Assign observations to the nearest centroid
- Recalculate centroids
- Repeat until convergence

Before training the model, the optimal number of clusters must be determined.

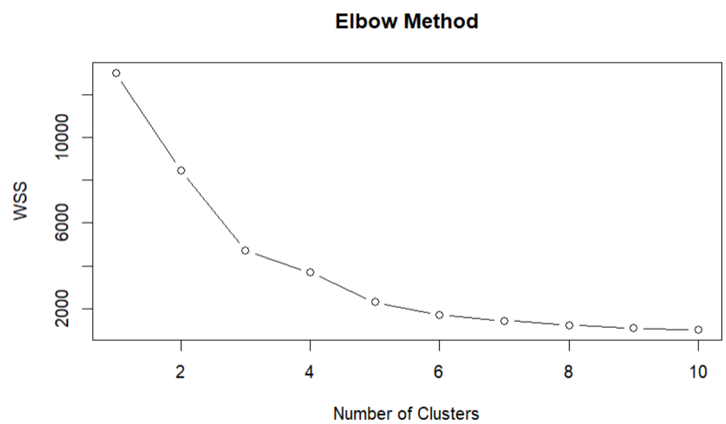

Figure 8: Elbow Method for selecting the optimal number of clusters

5.2 Elbow Method

The Elbow Method is used to identify the optimal value of k.

It analyzes the Within-Cluster Sum of Squares (WSS) for different values of k.

The graph shows a significant decrease in WSS until approximately $k = 3$, after which the improvement becomes smaller.

Therefore, $k = 3$ is selected as the optimal number of clusters for this project. The final K-Means model is trained using 3 clusters.

5.3 K-Means Cluster Interpretation

The obtained clusters can be interpreted as follows:

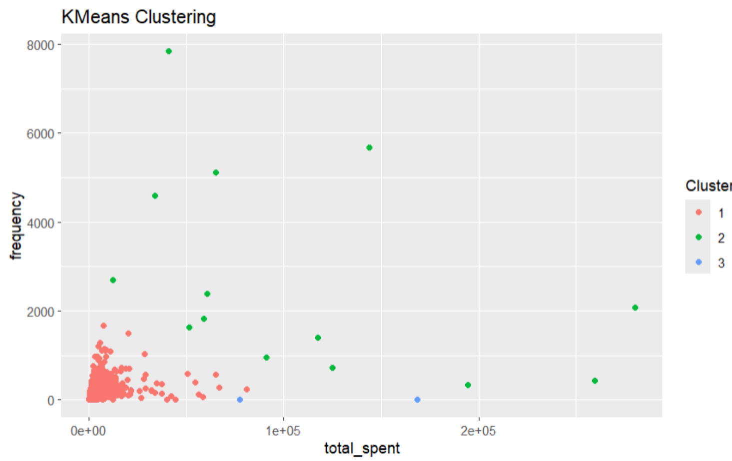

Figure 9: K-Means clustering visualization

The visualization shows the existence of different customer groups according to purchasing behavior.

- Cluster 1: Customers with low spending and low purchase frequency
- Cluster 2: Customers with moderate purchasing behavior
- Cluster 3: Customers with high spending and frequent purchases

Some customers correspond to high-value buyers with strong purchasing activity, while others represent customers with lower activity levels.

This segmentation helps identify customer categories that may require different business strategies.

5.4 Hierarchical Clustering

Hierarchical Clustering is another clustering approach that progressively groups observations according to their similarity.

Unlike K-Means, Hierarchical Clustering does not require specifying the number of clusters at the beginning.

The first step consists of computing the distance matrix between customers using normalized variables.

Then, the Ward method is applied in order to minimize variance within clusters.

5.5 Dendrogram Analysis

The hierarchical clustering process is visualized using a dendrogram.

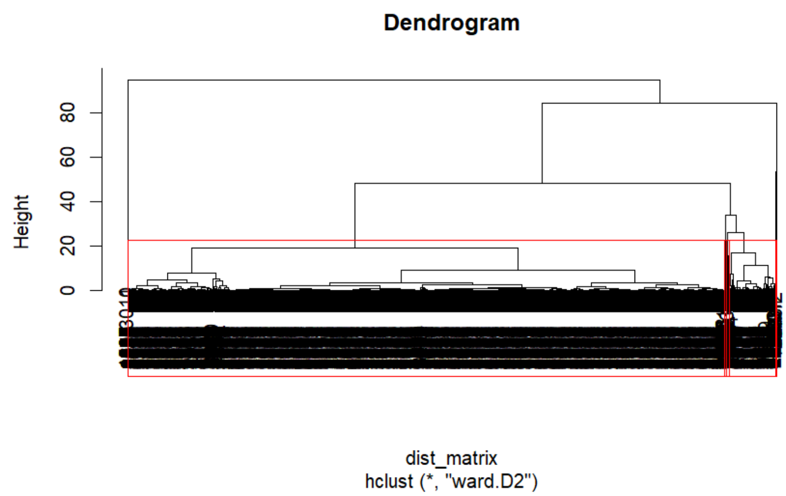

Figure 10: Hierarchical clustering dendrogram

The dendrogram illustrates how observations are progressively merged into larger groups.

The vertical axis represents the distance between clusters.

A horizontal cut of the dendrogram allows the selection of the final number of clusters.

In this project, the dendrogram suggests that $k = 3$ provides a meaningful segmentation of customers.

5.6 Hierarchical Clustering Visualization

The visualization confirms the presence of distinct customer groups.

Customers within the same cluster share similar purchasing patterns, while customers belonging to different clusters exhibit different behaviors.

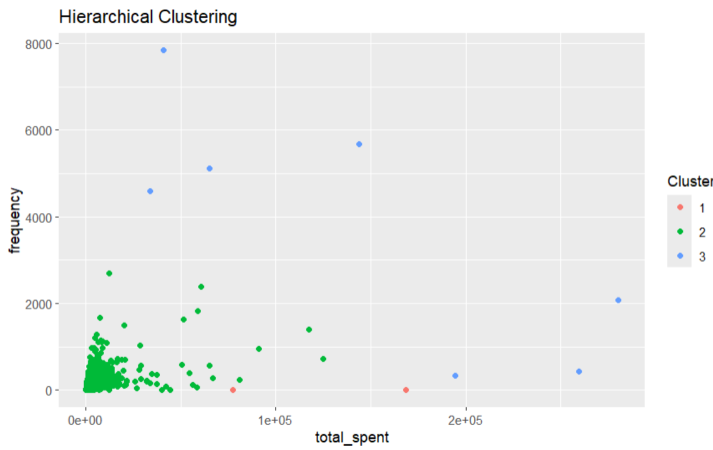

Figure 11: Hierarchical clustering visualization

5.7 Comparison Between Clustering Methods

Both clustering algorithms successfully identify meaningful customer segments.

However, several differences can be observed between the two approaches.

5.7.1 K-Means

Advantages:

- Simple and computationally efficient
- Suitable for large datasets
- Easy to interpret

Limitations:

- Requires choosing k in advance
- Sensitive to initialization
- Sensitive to outliers

5.7.2 Hierarchical Clustering

Advantages:

- Does not initially require a predefined number of clusters
- Provides a dendrogram for visualization
- Helps understand the structure of the data

Limitations:

- Computationally expensive
- Less scalable for large datasets

Overall, both methods produce relatively similar segmentation results. However, K-Means provides clearer cluster separation and better scalability for large retail datasets.

6 Conclusion

In this project, we applied two unsupervised learning algorithms, K-Means and Hierarchical Clustering, to perform customer segmentation using the Online Retail dataset.

The objective was to identify groups of customers with similar purchasing behaviors based on transactional data.

Several preprocessing steps were performed before clustering, including data cleaning, feature engineering, customer aggregation, and normalization.

The Elbow Method and dendrogram analysis both suggested that three clusters provide an appropriate segmentation of customers.

The results demonstrate that clustering techniques can effectively reveal hidden structures in customer behavior data.

K-Means provided efficient and interpretable clusters, while Hierarchical Clustering offered additional insights through dendrogram visualization.

These clustering methods can help businesses improve customer targeting, marketing strategies, and decision-making processes.

6.1 Limitations

This study has some limitations:

- Only a limited number of customer features were used
- No advanced dimensionality reduction techniques were applied
- Cluster quality metrics such as silhouette score were not included

These limitations may affect the precision of the customer segmentation.

6.2 Future Work

Several improvements can be considered for future work:

- Include additional customer behavioral variables
- Apply advanced clustering methods such as DBSCAN
- Use dimensionality reduction techniques such as PCA
- Evaluate clusters using silhouette analysis

7 Appendix

The complete R Markdown file and source code used in this project are provided separately.